Statistical Thinking for Data Science

Week 12: Bayesian thinking, and our own data

Belief, updating, and a study we ran ourselves

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First Semester, 2026/2027

Week 12: Bayesian thinking, and our own data

Learning Objectives

By the end of this week you should be able to:

- state probability as a degree of belief, and say when that reading is the useful one;
- put a prior on a proportion, update it with data, and report a credible interval;
- explain the difference between a credible interval and a confidence interval, in one sentence each;
- show how the prior stops mattering as data accumulate, and when it does not;
- run a small study end to end: questionnaire, anonymity, reliability, structure, and a regression on data you collected;
- say what your own class data can and cannot support.

12.1 The other reading of probability

Week 7 gave two readings of a probability: a long run frequency, and a degree of belief. Everything since has used the first. The second is needed whenever the thing you are uncertain about does not repeat: this deposit, this bridge, this semester's cohort. There is no long run of *this* ore body.

Bayesian inference applies Week 8's rule to parameters instead of patients:

$$\underbrace{P(\theta \mid \text{data})}_{\text{posterior}} \propto \underbrace{P(\text{data} \mid \theta)}_{\text{likelihood}} \times \underbrace{P(\theta)}_{\text{prior}}.$$

You state what you believed, you state how the data arise, and the rule tells you what to believe afterwards. The output is a distribution over the parameter, which is what most people wrongly think a confidence interval is.

12.2 Updating a proportion

For a proportion, the convenient prior is a Beta distribution, $\text{Beta}(a, b)$, which behaves like having already seen $a-1$ successes and $b-1$ failures. After x successes in n trials the posterior is

$$\text{Beta}(a + x,\ b + n - x),$$

which is why Beta and binomial are taught together: the updating is addition.

Example 1: how many students use the course tutor?

Of 200 students surveyed, 84 said yes.

Solution

With a weak prior $\text{Beta}(2, 2)$, centred on a half and worth about two observations, the posterior is $\text{Beta}(86, 118)$: mean 0.422, and a 95% credible interval of $(0.355,\ 0.490)$.

The frequentist interval from Week 9 was $(0.352,\ 0.488)$. With 200 observations and a weak prior, the two agree to within a few thousandths, as they almost always do. The difference is what you are allowed to say.

What you may say

Credible interval: "given the data and my prior, there is a 95% probability that the true proportion lies between 0.355 and 0.490." That is a statement about the parameter.

Confidence interval: "this procedure captures the truth 95% of the time in repeated sampling." That is a statement about the procedure.
The Bayesian sentence is the one people instinctively want. The price is that you had to declare a prior.

12.2.1 Does the prior take over?

Prior	Posterior	Mean	95% credible interval
Beta$(1,1)$, flat	Beta$(85,117)$	0.421	(0.354, 0.489)
Beta$(2,2)$, weak	Beta$(86,118)$	0.422	(0.355, 0.490)
Beta$(20,5)$, strongly optimistic	Beta$(104,121)$	0.462	(0.398, 0.528)

With 200 observations, a prior worth two observations is invisible and a prior worth 25 shifts the answer by four percentage points. Now shrink the data to 3 successes in 10:

Prior	Posterior	Mean	95% credible interval
Beta$(1,1)$	Beta$(4,8)$	0.333	(0.109, 0.610)
Beta$(20,5)$	Beta$(23,12)$	0.657	(0.495, 0.803)

Same data, opposite conclusions. The lesson is not that priors are dangerous; it is that with little data your conclusion depends on what you brought with you, whether or not you wrote it down. A frequentist analysis of 3 out of 10 is also nearly uninformative; it merely hides the fact behind a wide interval.

Report the sensitivity, not just the answer

Any Bayesian analysis worth reading states the prior, justifies it, and shows the answer under at least one competing prior. If a reviewer's reasonable prior reverses your conclusion, that is the finding: the data are too thin to settle the question.

12.2.2 Yesterday's posterior is today's prior

Updating is sequential and order does not matter. Take the first 50 responses, 21 of them yes: starting from Beta$(2,2)$ the posterior is Beta$(23,31)$, mean 0.426. Feed in the remaining 150, of which 63 are yes, and you reach Beta$(86,118)$: exactly the same place as analysing all 200 at once. This is why Bayesian monitoring of an ongoing study does not carry the penalty that repeated significance testing does in Week 10.

12.2.3 Answering the question that was asked

A Bayesian analysis can answer "what is the probability that more than half the class uses the tutor?" directly: it is the posterior probability above 0.5, which here is 0.012. No null hypothesis, no p-value, no double negative. Whether that is worth the extra assumptions is a judgement, and reasonable statisticians differ. Know both languages and say which one you are speaking.

12.3 A study we ran ourselves

For the rest of the course we work on data this class produced: the anonymous wellbeing questionnaire you completed in week 2. Every idea in the course appears in it.

12.3.1 Design and ethics, first

- The questionnaire is **anonymous**: no names, no index numbers, no email addresses, and no field that could identify one person by combination.
- It was voluntary, and skipping it costs nothing.
- It is a **teaching dataset**. It was collected for this classroom and has no ethics approval for research, so it is not published, not placed in a repository, and not written up for a journal or a conference. Analyses stay inside the course.
- We never report a cell with a handful of students in it. A result broken down by programme and gender can identify somebody even without a name.

The rule that outranks the statistics

Consent covers a purpose, not just a dataset. Data collected for teaching stay in teaching, however interesting the finding. Permission to analyse is not permission to publish.

12.3.2 Measurement: are the questions measuring one thing?

The instrument uses several items per scale, because one question is noisy. Two checks:

Reliability. Cronbach's alpha summarises how consistently the items in a scale move together:

$$\alpha = \frac{k\,\bar{r}}{1 + (k-1)\bar{r}},$$

with k items and $\bar{r}$ their average inter-item correlation. Seven items with an average correlation of 0.35 give $\alpha = 0.79$, which is usable; the same seven items at 0.20 give 0.64, which is not. Note that α rises with the number of items: 21 items at an average correlation of 0.20 reach 0.84 without the items being any better. A high alpha means consistency, not validity: twenty questions all asking the same thing badly will look excellent.

Structure. If three subscales exist, items within a subscale should correlate more with each other than with items from other subscales. A correlation matrix printed as a heatmap shows this immediately, and it is Week 6's advice applied to Week 5's statistic.

12.3.3 The analysis, week by week

Week	What it contributes to the case study
2 and 3	what question the survey answers, who was in the frame, who did not reply
4	distribution of each scale score, with shape and outliers, not just a mean
5	relationships between scales, and between a scale and study hours
6	the figures: distributions shown, no bar of means, greyscale safe
7 and 8	the probability model behind any claim, and conditioning on subgroups
9	intervals for every mean and proportion reported
10	the single pre-specified comparison, stated before we looked
11	a regression of one scale on study hours, adjusted for programme
12	a Bayesian version of the headline estimate, with a prior sensitivity check

12.3.4 What this dataset cannot support

- **Diagnosis.** Scale scores are not clinical assessments, and nobody in this class is qualified to use them that way.
- **Causation.** It is a one-off voluntary survey, so associations are associations. Studying more and feeling better may both follow from something else entirely.
- **Generalisation.** It describes the students who answered, in this class, this semester. Not UMaT, not Ghanaian students, not engineering students.

- **A hunt for significance.** With dozens of items, 5% of comparisons will be significant by chance. We state our comparison in advance and report the rest as exploratory.

12.4 Doing it on a computer

A Beta posterior needs no special software: `qbeta(c(0.025, 0.975), 86, 118)` in R, or `scipy.stats.beta(86, 118).ppf([0.025, 0.975])` in Python. For models beyond this, R has `brms` and `rstanarm`, and Python has `PyMC`, all of which sample the posterior rather than solving for it. The conceptual step is the same one you took on paper this week.

12.5 Glossary

Prior	your belief about a parameter before the data
Likelihood	how probable the data are for each value of the parameter
Posterior	your belief after the data
Conjugate prior	a prior whose family survives the update, as Beta does for a proportion
Credible interval	an interval containing the parameter with stated probability
Prior sensitivity	how much the conclusion changes under a different reasonable prior
Cronbach's alpha	a measure of internal consistency among items in a scale

12.6 Summary

Week 12 in six lines

1. Probability as belief lets you talk about one-off events, including parameters.
2. Posterior is proportional to likelihood times prior; for a proportion the update is addition.
3. A credible interval is about the parameter; a confidence interval is about the procedure.
4. With plenty of data the prior fades; with little data it decides, so report sensitivity.
5. Our class data is a teaching dataset: analysed here, published nowhere.
6. A study is only as good as its design, and no analysis in this course can repair a bad one.

Exercises

1. With a Beta$(2, 2)$ prior and 84 successes in 200, write down the posterior and compute its mean. Compare it with $\hat{p} = 0.42$ and explain the small difference.

2. Repeat with a Beta$(20, 5)$ prior. How far does the posterior mean move, and would you defend that prior to a sceptical reader?

3. Now suppose only 10 students were surveyed, 3 of whom said yes. Compute the posterior under both priors and comment on what the comparison shows.

4. Write one sentence interpreting a 95% credible interval, and one interpreting a 95% confidence interval, for the same estimate. Make the difference unmistakable.

5. Update sequentially: first 50 responses with 21 yes, then 150 more with 63 yes. Show that you reach the same posterior as analysing all 200 at once, and say why this matters for monitoring a study.

6. A seven item scale has an average inter-item correlation of 0.35. Compute Cronbach's alpha. What happens to alpha if you pad the scale with seven more items of the same quality, and why is that not necessarily an improvement?

7. Using the class survey, state one question the data can answer and one it cannot, and justify both in two sentences each.

8. Write the paragraph of your project report that states what your data cannot support. Be specific, and resist the temptation to be modest in general terms.

Answers to the odd-numbered exercises

1. Posterior Beta$(86, 118)$, mean $86/204 = 0.4216$, against $\hat{p} = 0.42$. The prior adds two pseudo-observations, one success and one failure, which pulls the estimate a fraction towards 0.5; with 200 real observations the pull is invisible.

3. Flat prior: Beta$(4, 8)$, mean 0.333, interval $(0.11, 0.61)$. Optimistic prior: Beta$(23, 12)$, mean 0.657, interval $(0.49, 0.80)$. Same data, opposite conclusions, because ten observations cannot outweigh a prior worth twenty-five. With data this thin, report the sensitivity or collect more.

5. After 50: Beta$(23, 31)$, mean 0.426. Adding 63 successes and 87 failures gives Beta$(86, 118)$, identical to the one-shot analysis. Updating is order independent, so looking at your data as it arrives does not inflate any error rate, unlike repeated significance testing.

7. Answerable: "what proportion of the students who responded report high stress in exam week?", because that is a description of the respondents, reportable with an interval. Not answerable: "does studying more cause lower stress?", because the survey is observational, one-off and voluntary, so study hours and stress are tangled with everything from programme to term-time work.

After the course

You now have the whole chain: a question, a design, data you can describe, pictures that do not lie, a probability model, an interval, a test, a model, and a way to say what you believe afterwards. The part that will serve you longest is none of the formulas. It is the habit of asking where the numbers came from, and of saying out loud what your analysis cannot support.